

SCOPE & SEQUENCE • GRADES K-5, WITH MIDDLE-SCHOOL EXTENSIONS

Kitchen STEM



Science, Technology, Engineering and Math through cooking, baking and food exploration

A grade-band progression across the four strands, a companion deck of thirty-eight activity cards, and the notes an adult needs to teach any of it well. Every card is written for both a classroom and a kitchen table.

Why the kitchen

A kitchen is the only lab most children have standing access to, and it is a remarkably good one. Every variable is adjustable, every result is immediate, and the feedback is not a grade — it is a texture, a smell, a collapse, a rise. A child who under-measures baking powder does not get a red mark; she gets a flat pancake, and she can see why.

That immediacy is what makes food such a strong carrier for STEM. Fractions in a workbook are abstract. Fractions in a doubled cookie recipe are load-bearing: get them wrong and the batch is wrong. Heat transfer is a diagram until the child touches the handle of a metal spoon left in soup and understands conduction in her hand. The kitchen collapses the distance between a concept and its consequence.

It is also unusually democratic. It requires no lab budget, it accommodates a huge range of ability and attention, and it produces something at the end that a child can hand to someone they love. That last part is not incidental — it is often what keeps a reluctant learner at the table for the second half of the lesson.

How this set is built

There are two pieces. This document is the scope and sequence: what concepts live in which grade band, organized by STEM strand, so you can see the arc and place any single activity inside it. The companion deck holds one card per activity — materials, steps, the questions worth asking, and the explanation the adult needs in order to answer the follow-up.

Every card carries two columns near the bottom: **In the classroom** and **At home**. The science does not change between them, but almost everything else does — group size, prep, substitutions, how much of the mess you are willing to absorb. Read the column that matches your situation and ignore the other one.

The cards are not a curriculum in sequence. They are a bank. Pull the four that match the unit you are already teaching, or the one that matches what is already in the refrigerator.

The adult's job

The most common way kitchen STEM fails is that the adult does the thinking out loud. It is very fast to say *“the baking soda reacts with the vinegar and makes carbon dioxide”* and very slow to let a seven-year-old notice that the bubbles only started when the two wet things met. The slow version is the lesson.

So each card gives you the explanation in a section called **What is actually happening**, and that section is for you, not for the child. Hold it. Use it to ask a better question, to know which observation matters, and to answer honestly when a child asks something you did not anticipate. Deliver it as an explanation only after they have made a prediction and watched it succeed or fail.

Two questions do most of the work at every grade: **“What do you think will happen?”** before, and **“What surprised you?”** after. A wrong prediction that a child stated out loud is worth more than a right answer they were handed.

Safety, and building judgment rather than rules

Heat and blades are the two real hazards, and both are better handled by graduated responsibility than by prohibition. A child who has never been allowed near a knife at eight is not safer at twelve; they are simply inexperienced at twelve.

A workable progression: at K–2, children tear, mash, stir cold things, measure, press, and cut soft foods with a table knife or a nylon lettuce knife. At grades 3–4 they add a small serrated knife with a bridge or claw grip on stable foods, and they operate the toaster oven and the microwave with an adult in the room. At grade 5 they move to a chef’s knife on a damp-towel-anchored board and manage a stovetop burner directly. The adult’s job shifts from doing to spotting.

Independent of age, name the hazard out loud before it is live: *“The pan handle is turned in so nobody catches it with a sleeve.”* *“Steam burns worse than water — lift the lid away from your face.”* Children who hear hazards named learn to name them, which is the skill that actually transfers out of the kitchen.

Standing rules worth keeping: hair tied, sleeves up, hands washed and re-washed after raw egg or raw meat, a damp towel under every cutting board, oven mitts dry (a wet mitt conducts heat straight through), and nobody carries something hot across a room where other people are moving.

Allergies, restriction and inclusion

Before any of this, know your room. The common allergens in these activities are milk, egg, wheat, soy, sesame, peanut and tree nut; several cards also involve raw flour, which carries its own risk and should not be tasted uncooked.

Design the activity so that the child with the restriction is doing the same science, not a diminished version of it. Emulsification works with aquafaba as well as egg yolk. Leavening works in a gluten-free batter. A child who cannot taste the result can still run the experiment, record the data, and be the one who reports out — but do that by giving them a real role, not by handing them a clipboard as a consolation.

For children who are sensory-sensitive, the wet-hands steps (dough, mashing, batter) are the ones to make optional first, and a spoon or a glove usually solves it. For children who need predictability, the recipe-as-algorithm framing on card E01 is genuinely useful: the numbered card *is* the visual schedule.

A note on the standards codes

Each card lists standards *touchpoints* — NGSS performance expectations for science and engineering, Common Core for math (and ELA where a card turns on reading a procedural text), ISTE for the technology strand. They are honest pointers to where the activity lives in the framework, not claims that a single forty-minute activity fully meets a performance expectation. Almost none do on their own; performance expectations are designed to be met across a unit.

If you are in a state with its own adopted standards, treat these as a crosswalk starting point and map them across once. The underlying concepts are stable even where the codes are not.

Kindergarten – Grade 2

Notice, sort, describe, predict

At this band the kitchen is a place to build vocabulary and careful noticing. Children are not yet controlling variables; they are learning that things have properties, that properties can be compared, and that some changes go backward (melted butter re-solidifies) while others do not (a scrambled egg stays scrambled). That single distinction — reversible versus irreversible — is the spine of the band, and it is explicitly a second-grade expectation. Mathematically the work is counting, direct comparison, and non-standard then standard measurement. Keep the sessions short, the steps few, and the talk constant.

WHAT PROGRESS LOOKS LIKE

- Uses comparison language without prompting: thicker, colder, rougher, faster.
- Makes a prediction before acting, even a wrong one, and remembers making it.
- Sorts the same set of foods two different ways and can say what rule they used.
- Names one change that can be undone and one that cannot.
- Reads a measuring cup to the line rather than filling it to the top.

CONCEPTS BY STRAND • SCIENCE

SCIENCE

- Properties of materials: hard/soft, rough/smooth, liquid/solid, floats/sinks.
- States of matter and phase change — melting, freezing, boiling, steam.
- Reversible vs. irreversible change (melted chocolate vs. a cooked egg).
- Mixtures and solutions at the noticing level: what disappears in water, what does not.
- Plant parts we eat — root, stem, leaf, flower, fruit, seed.
- Seeds sprout; plants need water and light.
- The five senses, and the discovery that smell does most of the work of taste.
- Living things can be too small to see, and some of them make our food.
- Hot moves to cold; some things keep heat in.

CONCEPTS BY STRAND • TECHNOLOGY, ENGINEERING, MATHEMATICS

TECHNOLOGY

- A tool is something a person made to do a job better than hands can.
- Matching tool to task: whisk vs. spoon, grater vs. knife, sieve vs. hands.
- Measuring instruments — the cup, the spoon, the scale, the timer.
- Reading a thermometer as a number that goes up and down.
- Food does not appear in the store: a first pass at where it comes from.
- Ways people keep food from spoiling — cold, dry, sealed.

ENGINEERING

- A recipe is a set of steps in an order, and the order matters.
- Building with edible materials: what makes a tower stand up.
- Testing an idea, and trying a second version when the first one fails.
- Simple containers — what keeps something safe, cold, or dry.

MATH

- Counting, one-to-one correspondence, and equal shares.
- Comparing measurable attributes directly — heavier, taller, more.
- Halves and quarters as fair shares of a real object (K.G, 1.G, 2.G).
- Filling to a line: standard units of volume and weight.
- Sorting into categories and building a picture or bar graph of the result.
- Sequencing and elapsed time in whole minutes.
- Shapes in food, and symmetry when you cut things in half.

CARDS FOR THIS BAND

These 17 cards from the deck are written to be run at kindergarten – grade 2. Several appear in more than one band and are simply run at a different level of demand — the card's **Extend it** line says how.

S01 Bubbles on Demand	S02 Something in the Jar Is Alive	S06 The Cabbage That Tells You	S08 Where Did the Sugar Go?
S10 Which Part Are We Eating?	S13 From Seed to Something You Eat	S14 Taste Is Mostly Smell	S16 The Floating Egg
S17 How Slow Does It Pour?	S18 Keep It Cold, Keep It Hot	S20 The End of the Meal Is Not the End	T01 The Machines in the Drawer
E01 A Recipe Is an Algorithm	E03 Load-Bearing Gingerbread	E06 Get It There in One Piece	M01 Fractions That Actually Matter
M08 Run a Real Taste Test			

If you only do four

- S10 **Which Part Are We Eating?** Needs no heat, no equipment and no preparation beyond a grocery run, and it reorganizes how a child sees every vegetable.
- S01 **Bubbles on Demand** The clearest chemical change available at this age, and it produces breakfast.
- S16 **The Floating Egg** One surprising result, one real household use, and a concept the child can test again next week.
- M01 **Fractions That Actually Matter** Start here if the band's math is the priority. Practice with dry rice costs nothing.

Grades 3 – 5

Control a variable, measure it, explain it

This is the band where kitchen STEM does its heaviest lifting. Children can now hold one thing constant while changing another, which converts cooking from a procedure into an experiment. It is also the band where fractions, ratio and measurement conversion arrive in the math curriculum, and a measuring cup is a better manipulative than anything in the supply closet — partly because the fractions are consequential and partly because ounces of weight and fluid ounces of volume sit side by side on the counter, which is where most people's misconception about them can finally be broken. Chemically, this is where leavening, browning, emulsions and denaturation belong, and where the engineering design cycle can be run honestly across multiple batches.

WHAT PROGRESS LOOKS LIKE

- Changes exactly one thing between two batches and can say why that matters.
- Distinguishes a physical change from a chemical one using evidence, not vocabulary.
- Scales a recipe up or down and checks the result against an estimate.
- Explains a result in terms of a cause they cannot see — gas, protein, particles.
- Redesigns after a failure and names what the failure told them.
- Weighs rather than scoops when precision matters, and can say why.

CONCEPTS BY STRAND • SCIENCE

SCIENCE

- Chemical vs. physical change, with evidence: new gas, new color, new smell, not reversible.
- Chemical leavening — base plus acid releases carbon dioxide.
- Biological leavening — yeast is alive, eats sugar, exhales gas.
- Maillard browning and caramelization as two different browning routes.
- Protein denaturation: eggs setting, whites whipping, gluten developing.
- Emulsions, and the molecules that hold oil and water together.
- Acids and bases, and using a plant pigment as a pH indicator.
- Crystallization, and the idea that crystal size is controllable.
- Solutions, suspensions and colloids; solubility changes with temperature.
- Starch gelatinization — why a sauce thickens.
- Fermentation and the microbiology of yogurt and sourdough.
- Enzymes as biological scissors (raw pineapple vs. gelatin).
- Heat transfer by conduction, convection and radiation.
- Density, buoyancy, and layering liquids.
- Viscosity, and how temperature changes it.
- Decomposers and the cycling of matter through compost.
- Where food grows, and why climate and season decide it.

CONCEPTS BY STRAND • TECHNOLOGY, ENGINEERING, MATHEMATICS

TECHNOLOGY

- Simple machines hidden inside kitchen tools — lever, wedge, screw, wheel-and-axle.
- Instruments and calibration: testing a thermometer against a known point.
- Precision vs. accuracy, in the hands, with a scale.
- Preservation technologies compared: canning, freezing, drying, fermenting, pasteurizing.
- Spreadsheets as a tool for costing a recipe per serving.
- Writing a tiny unit-converter program; algorithms as recipes and recipes as algorithms.
- Data logging — recording temperature over time and plotting it.

ENGINEERING

- The full design cycle: define, plan, build, test, improve.
- Controlled-variable testing across batches — the cookie experiment.
- Structural engineering with edible materials: loads, joints, failure modes.
- Process engineering: assembly lines, bottlenecks, throughput.
- Thermal design: insulation, solar ovens, keeping things hot or cold on purpose.
- Packaging and protective design against a defined force.
- Criteria and constraints stated before building, not after.

MATH

- Fractions: equivalence, addition with unlike denominators, multiplication by a whole number.
- Ratio and proportional reasoning — 3:1 vinaigrette, brines, baker's percentages.
- Weight vs. volume, and ounces vs. fluid ounces.
- Unit conversion within and between systems.
- Area, perimeter and circumference applied to pans; volume of a rectangular pan.
- Surface-area-to-volume reasoning and why small things cook faster.
- Elapsed time and backward planning from a fixed finish time.
- Unit price, cost per serving, and comparing package sizes.
- Collecting categorical and rating data, then graphing and summarizing it.
- Estimation and reasonableness checks on every scaled recipe.

CARDS FOR THIS BAND

These 38 cards from the deck are written to be run at grades 3 – 5. Several appear in more than one band and are simply run at a different level of demand — the card's **Extend it** line says how.

S01 Bubbles on Demand

S02 Something in the Jar Is Alive

S03 Why Brown Tastes Better

S04 Rewriting a Protein

S05 Making Oil and Water Agree

S06 The Cabbage That Tells You

S07 Big Crystals, Small Crystals

S08 Where Did the Sugar Go?

S09 How a Sauce Gets Thick

S10 Which Part Are We Eating?

S11 Farming Microbes

S12 The Jello That Refuses to Set

S13 From Seed to Something You Eat

S14 Taste Is Mostly Smell

S15 Three Ways Heat Travels

S16 The Floating Egg

S17 How Slow Does It Pour?

S18 Keep It Cold, Keep It Hot

S19 Where Does This Actually Come From?

S20 The End of the Meal Is Not the End

T01 The Machines in the Drawer

T02 Is the Thermometer Telling the Truth?

T03 Five Ways to Cheat Time

T04 The Kitchen Goes Digital

E01 A Recipe Is an Algorithm

E02 The Cookie Experiment

E03 Load-Bearing Gingerbread

E04 Find the Bottleneck

E05 Cooking with the Sun

E06 Get It There in One Piece

M01 Fractions That Actually Matter

M02 Cooking Without a Recipe

M03 An Ounce Is Not an Ounce

M04 The Pan Problem

M05 Why Small Things Cook Faster

M06 Dinner at Six

M07 What Does This Actually Cost?

M08 Run a Real Taste Test

If you only do four

- E02 **The Cookie Experiment** The single best use of a kitchen at this band: a real controlled-variable test that takes seventy-five minutes and teaches method rather than content.
- S06 **The Cabbage That Tells You** Cheap, visual, repeatable all week, and it makes an invisible property visible.
- M03 **An Ounce Is Not an Ounce** Four minutes of it — five children scooping flour — breaks a misconception most adults still hold.
- S15 **Three Ways Heat Travels** Explains a dozen everyday kitchen instructions the child has been following without knowing why.

Grades 6 – 8 extension

Quantify the mechanism, argue from evidence

Included because most of these activities extend cleanly upward rather than needing replacement. What changes at middle school is the demand for quantification and for argument. A fifth grader observes that the sourdough rose faster in the warm spot; a seventh grader plots rise height against temperature, identifies the optimum, and argues from the graph about enzyme and microbial activity. Ratio reasoning becomes proportional reasoning with a constant of proportionality; browning becomes a named reaction between amino acids and reducing sugars; heat transfer becomes something they calculate rather than describe.

WHAT PROGRESS LOOKS LIKE

- States a hypothesis in terms of a mechanism, not just an outcome.
- Plots two variables and reads a relationship off the graph.
- Uses a constant of proportionality to scale rather than repeated addition.
- Argues from their own data when it disagrees with the recipe.
- Identifies a confound in someone else's experimental design.

CONCEPTS BY STRAND • SCIENCE

SCIENCE

- Chemical reactions evidenced by property change before and after (MS-PS1-2).
- Thermal energy, particle motion and state change quantified (MS-PS1-4, MS-PS3-4).
- Maillard chemistry named: amino acids plus reducing sugars, and why pH and water activity matter.
- Denaturation and coagulation temperatures measured, not estimated.
- Microbial growth curves; temperature as the controlling variable.
- Enzyme specificity and denaturation by heat (canned vs. fresh pineapple).
- Osmosis and diffusion — brining, salting, and why cucumbers weep.
- Energy transfer efficiency in a solar oven.

CONCEPTS BY STRAND • TECHNOLOGY, ENGINEERING, MATHEMATICS

TECHNOLOGY

- Instrument error, tolerance, and repeated measurement.
- Spreadsheet modeling: cost, yield, scaling, and break-even on a bake sale.
- Programming a converter or a scaling calculator with real input validation.
- Food systems, supply chains, labels and traceability.

ENGINEERING

- Optimization against competing criteria (cost vs. quality vs. time).
- Systematic testing across a factor grid rather than one change at a time.
- Thermal system design with measured performance (MS-PS3-3).
- Failure analysis and iteration documented as an engineering notebook.

MATH

- Ratios, rates, unit rates and percentage (6.RP).
- Proportional relationships and constants of proportionality (7.RP).
- Circumference, area, surface area and volume in pan and portion problems (7.G).
- Statistical variability: mean, median, spread, and outliers in taste-test data (6.SP).
- Scale drawings and dimensional scaling — why doubling a cake pan is not doubling a cake.

CARDS FOR THIS BAND

Every card in the deck carries an **Extend it** line written for this band. The extensions below are where each strand goes when the demand shifts to quantification and argument.

S01 Bubbles on Demand

S02 Something in the Jar Is Alive

S03 Why Brown Tastes Better

S04 Rewriting a Protein

S05 Making Oil and Water Agree

S06 The Cabbage That Tells You

S07 Big Crystals, Small Crystals

S08 Where Did the Sugar Go?

S09 How a Sauce Gets Thick

S10 Which Part Are We Eating?

S11 Farming Microbes

S12 The Jello That Refuses to Set

S13 From Seed to Something You Eat

S14 Taste Is Mostly Smell

S15 Three Ways Heat Travels

S16 The Floating Egg

S17 How Slow Does It Pour?

S18 Keep It Cold, Keep It Hot

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M01 Fractions That Actually Matter

M02 Cooking Without a Recipe

M03 An Ounce Is Not an Ounce

M04 The Pan Problem

M05 Why Small Things Cook Faster

M06 Dinner at Six

M07 What Does This Actually Cost?

M08 Run a Real Taste Test

If you only do four

- M02 **Cooking Without a Recipe** Baker's percentages are proportional reasoning with a constant of proportionality, arriving in a form professionals actually use.
- S11 **Farming Microbes** Run with daily pH measurement it becomes a genuine microbiology investigation with a curve to argue from.
- E05 **Cooking with the Sun** Four mechanisms at once, a measurable output, and a design competition with a real constraint.
- S03 **Why Brown Tastes Better** Where the chemistry can finally be named — amino acids and reducing sugars — rather than described.

The thread underneath

Cooking carries social and emotional learning whether or not anyone plans for it, so it is worth planning for. Each card names one thread explicitly, but four recur:

Tolerating a failed batch. Food that does not work is a uniquely safe failure — low stakes, visible, and fixable in twenty minutes. Naming it as data rather than as a mistake is a stance children carry into work that is not safe.

Interdependence. An assembly line fails at its slowest station, which means the fast children cannot succeed alone. Few classroom structures make that as unarguable.

Delayed gratification with a visible clock. Dough that needs ninety minutes teaches waiting differently than a timer on a screen does, because the child can see the change accumulating.

Feeding someone. The end of most of these activities is handing food to another person. That is a small act of care with a real recipient, and it is the reason a child who will not finish a worksheet will finish a batch.

Equipment, pantry and substitutions

Almost everything in the deck runs on ordinary groceries. This is the short list of what is worth having on hand before you start, and the swaps that keep the science intact when an ingredient is off the table.

EQUIPMENT WORTH OWNING

- A digital scale that reads grams and switches to ounces — the single highest-value purchase in this whole set, because it makes ratio work honest.
- An instant-read thermometer, ideally two so children can compare readings.
- Nested dry measuring cups and a clear liquid measuring cup — you need both, and the difference between them is itself a lesson.
- Two identical sheet pans, so paired batches bake under identical conditions.
- A few clear glass jars or cups — almost every observation activity improves when the children can see the side of the container.
- Parchment paper, a bench scraper, a fine sieve, a whisk, and a nylon or plastic lettuce knife for young hands.

PANTRY STAPLES THESE CARDS DRAW ON

- Flour (all-purpose), sugar (granulated and brown), salt, baking soda, baking powder, active dry or instant yeast.
- Eggs, butter, milk, a neutral oil, white vinegar, lemons.
- Cornstarch, honey or corn syrup, unflavored gelatin, plain yogurt with live cultures.
- One red cabbage (the pH indicator), a fresh pineapple and a can of pineapple, food coloring.
- A produce spread that covers every plant part: carrot or radish, asparagus or kohlrabi, lettuce or spinach, broccoli or cauliflower, apple or tomato, sunflower or pumpkin seeds.

SUBSTITUTIONS THAT PRESERVE THE SCIENCE

- Egg-free emulsion: aquafaba (chickpea liquid) or a teaspoon of mustard as the emulsifier.
- Dairy-free: any plant milk works for leavening and gelatinization; for buttermilk, add a tablespoon of vinegar to a cup of plant milk.
- Gluten-free: leavening, browning and gelatinization all behave normally; only the gluten development card needs wheat, and it can be run as a demonstration.
- Gelatin-free: agar sets without animal product, but note it is *not* affected by pineapple enzyme, which changes the result on that card — use it as the control.
- No oven: nearly every chemistry card runs in a skillet, a toaster oven or a microwave.

Five ways to assess without a test

None of these require grading a batch of food. They assess reasoning, which is the thing actually being built.

Prediction slips

Before the activity, every child writes one sentence: *I think ___ will happen because ___*. Collect them. The *because* half is the assessment; the prediction itself is not the point. Hand them back at the end and ask what they would change.

Two-column observation

A sheet split into **What I saw** and **What I think caused it**. Children at every band can do this, and the widening gap between the columns over a year is a legible record of growth.

Explain it to someone smaller

Ask the child to teach the concept to a younger sibling or a lower grade. Compression without distortion is a strong signal of real understanding, and it exposes memorized vocabulary immediately.

The redesign paragraph

After any engineering card: *My first design failed because ___. I changed ___. The result was ___*. This is the whole design cycle in three blanks and it is the most transferable artifact in the set.

Recipe rewrite

Give a recipe for four and ask for it rewritten for ten, with the work shown. It assesses fractions, ratio, unit conversion and reasonableness in one page.

Index of cards

Thirty-eight cards. The companion deck holds one per page, in this order.

Science · Chemistry

S01 Bubbles on Demand

Chemical leavening: base + acid – carbon dioxide

S02 Something in the Jar Is Alive

Biological leavening: yeast, sugar and fermentation

S03 Why Brown Tastes Better

Maillard browning vs. caramelization

S04 Rewriting a Protein

Denaturation — eggs, foam and gluten

S05 Making Oil and Water Agree

Emulsification

S06 The Cabbage That Tells You

Acids, bases and a pH indicator you can eat

S07 Big Crystals, Small Crystals

Crystallization and why texture is controllable

S08 Where Did the Sugar Go?

Solutions, suspensions, colloids, and solubility

S09 How a Sauce Gets Thick

Starch gelatinization

Science · Biology

S10 Which Part Are We Eating?

Plant anatomy on a plate

S11 Farming Microbes

Fermentation — yogurt, sourdough and sauerkraut

S12 The Jello That Refuses to Set

Enzymes as biological scissors

S13 From Seed to Something You Eat

Germination, and the food in your pantry that is still alive

S14 Taste Is Mostly Smell

Sensory science, and testing a claim about your own body

Science · Physics

S15 Three Ways Heat Travels

Conduction, convection and radiation

S16 The Floating Egg

Density, buoyancy and layering

S17 How Slow Does It Pour?

Viscosity, and what temperature does to it

S18 Keep It Cold, Keep It Hot

Insulation and thermal design

Science · Earth & Environment

S19 Where Does This Actually Come From?

Seasons, climate and the map behind a meal

S20 The End of the Meal Is Not the End

Decomposers and the cycling of matter

Technology

T01 The Machines in the Drawer

Simple machines hiding in kitchen tools

T02 Is the Thermometer Telling the Truth?

Instruments, calibration and measurement error

T03 Five Ways to Cheat Time

Food preservation technologies compared

T04 The Kitchen Goes Digital

Spreadsheets, converters, and code that does the boring part

Engineering

E01 A Recipe Is an Algorithm

Sequence, dependency, and why order matters

E02 The Cookie Experiment

Controlled variables, run properly, across real batches

E03 Load-Bearing Gingerbread

Structures, joints and failure analysis

E04 Find the Bottleneck

Process engineering with an assembly line

E05 Cooking with the Sun

Solar oven design and energy capture

E06 Get It There in One Piece

Protective packaging against a defined force

Mathematics

M01 Fractions That Actually Matter

Halving, doubling, equivalence, and adding unlike denominators

M02 Cooking Without a Recipe

Ratios, proportions and baker's percentages

M03 An Ounce Is Not an Ounce

Weight vs. volume, and why bakers use a scale

M04 The Pan Problem

Area, circumference and volume where it counts

M05 Why Small Things Cook Faster

Surface area to volume, and why uniform cutting matters

M06 Dinner at Six

Elapsed time and backward planning

M07 What Does This Actually Cost?

Unit price, cost per serving, and comparing packages

M08 Run a Real Taste Test

Collecting, graphing and summarizing data you generated